1. Introduction

1.1. Purpose

The purpose of this project is to design a student Grade and GPA calculator system that helps student accurately calculate their GPA and see their grades each semester based on their course grades and credit hours The system aims to reduce manual calculation errors

1.2. Document Conventions

This document follows the IEEE 830 Software Requirements Specification (SRS) standard. The following conventions are used throughout the document:

- Headings and subheadings are numbered hierarchically to improve readability and organization
- Use cases are labeled using the format (Use case #) and are described with a clear structure including actor, description, and flow of events, preconditions and postconditions
- The term "system" refers to the Student Grade and GPA Calculator
- Input values provided by users (such as course name, marks, grades, and credit hours) are assumed to be entered through a user interface
- The keywords "shall", "should", and "may" are used with the following meanings:

  "Shall" indicates a mandatory requirement

  "Should" indicates a recommended requirement

  "May" indicates an optional feature

- GPA is represented using standard decimal notation

This document is intended to be clear, consistent, and unambiguous to ensure proper understanding by all stakeholders

1.3. Intended Audience and Reading Suggestions

Intended Audience:
Developers will use this document as a primary reference for implementing system functionalities according to the specified requirements

Project managers will use this document to plan, monitor, and control the development process, ensuring that the project stays aligned with its objectives and constraints

Reading Suggestions :

Students should read introductory guides on how GPA is calculated, time management handbooks to balance coursework, and their institution's academic standing policy to understand grading thresholds and consequences

Developers & System Designers are recommended to read about clean coding practices, UI/UX design for educational tools, database structure for student records, and accessibility guidelines to build a reliable and user-friendly system

1.4. Product Scope

In scope :

1. GPA Calculation

- Convert final grades to grade points based on a standard scale

  Factor in credit hours per course

- Compute both semester GPA

2 .Course Management

Input course details:

- Course name
- Credit hours
- Grade or marks

Out of scope :

1- It will not calculate accumulative GPA

2-Course registration or timetable management

3-Saving or storing student data permanently

4-Integration with any university system or database

1.5. References

1- Standard GPA Calculation System

- Most universities follow a 4.0 GPA scale
- Source: University grading policies (general academic standard)

2- Cairo University Academic Regulations

- Used as a reference for grading systems and credit hour structure

3- IEEE 830-1998 SRS Standard

- For writing the Software Requirements Specification document

4- SonarQube Documentation

- For code quality and clean code practices

5-Course Material

- Software Process & Quality Management lectures and lab materials

2. Overall Description

2.1. Product Perspective

The Student Grade & GPA Calculator is a new, self-contained software product designed to simplify and automate the process of calculating student grades and GPA in academic institutions

The system does not depend on any external university systems or databases. All data (courses, grades, and credit hours) is entered manually by the user

The application runs as a web application and it interacts only with the user through a graphical user interface

2.2. Product Features

1-Grade conversion

- The system allows the user to input a numeric mark
- The system converts the mark into a corresponding letter grade based on a predefined grading scale
- The system displays the resulting letter grade

2-Calculate GPA by grade :

- The system allows the user to enter multiple courses

  for each course, the user inputs:

- Course name
- Letter grade
- Credit hours

  The system calculates the GPA using the standard 4.0 scale
  System displays GPA

3-Calculate GPA by marks :

- The system allows the user to enter multiple courses
- For each course, the user inputs:
    - Course name
    - Course mark
    - Credit hours
- The system converts marks into letter grades
- The system calculates GPA based on credit hours and grade points
- The system displays the final GPA

2.3. User Classes and Characteristics

User Class 1: Students (Most Important)

Students use this system regularly every semester and after every major exam. Their technical expertise ranges from low to medium The features they rely on most are grade entry GPA checking and viewing their grades and GPA students are the highest priority user class and always come first in every design The interface must feel like a helpful friend not a complicated machine it must be simple

User Class 2: System Developers (Support Role)

Developers are the builders of this system They work behind the scenes to make sure everything functions correctly, securely, and efficiently

Developers engage with the system during the development phase, during updates and whenever maintenance is required Their technical expertise is high and they are fully comfortable working with code , APIs, and system architecture they have full access to the entire system including the backend logic and source code. They operate in a support role rather than as a primary user class

2.4. Operating Environment

The Student Grade and GPA Calculator is a standalone application that operates in a standard computing environment without requiring external systems or database integration

- Platform: The system runs on personal computers (PCs) or laptops.
- Operating System: Compatible with common operating systems such as Windows, macOS, and Linux
- User Interface: The system provides a simple user interface
- Hardware Requirements:

  Minimum 4 GB RAM

- Software Requirements:

  Any modern programming environment or runtime required to execute the application

2.5. Design and Implementation Constraints

The development of the Student Grade and GPA Calculator is subject to the following constraints:

- Data constraint

  All data entered by the user will be temporary and stored only during runtime

- No External System or Database Integration Constraint:
  The system shall not interface with any external systems and shall not use any database all data shall be entered by the user at runtime and processed in-memory only
- Version Control Constraint:
  The development process shall use Git and GitHub for source code management
- Code Quality Constraint:
  The code shall comply with quality standards enforced by tools such as SonarQube, including constraints on code duplication, maintainability, and absence of critical issues
- Continuous Integration Constraint:
  The system shall include a CI pipeline (GitHub Actions) that automatically builds and tests the application upon code updates.
- Performance Constraint

  The system should provide immediate calculation results with minimal delay

- **User Input Constraint:**
  The system design shall rely entirely on manual user input (course name, marks, grades, credit hours), with no automated data sources

2.6. User Documentation

Quick start guide:

Getting started with the Student Grade & GPA Calculator is simple:

1. Open the web app

2. Add all your courses

3. Choose what you want to do:

    - Calculate GPA using marks
    - Calculate GPA using letter grades
    - Convert mark to letter grade

4. Enter your course details:

    - Course name
    - Mark or letter grade
    - Credit hours

5. Click Calculate to see your GPA instantly.

2.7. Assumptions and Dependencies

Assumptions:

The following assumptions are made during the development of the Student Grade & GPA Calculator:

- Users have basic knowledge of how GPA systems work
- Users will enter correct and valid data
- The grading scale used in the system is fixed
- The system will run in a standard environment without requiring special configurations

Dependencies:

The system depends on the following external factors and components:

- Development Tools**:** The project depends on tools such as Git and GitHub for version control and collaboration
- Code Quality Tool: The use of SonarQube is required to analyze and ensure code quality
- Programming Environment: The system depends on the chosen programming language and its runtime environment
- User Device: The system depends on the availability of a user device

3. System Features

1-Grade conversion

This feature allows the user to convert numeric marks into corresponding letter grades based on a predefined grading scale

2- GPA Calculation using Marks

This feature enables the user to calculate GPA by entering course marks and credit hours

3- GPA Calculation using Letter Grades

This feature allows the user to calculate GPA using letter grades and credit hours

4- Course Data Entry

This feature allows users to input course information such as course name, marks or grades, and credit hours

4. External Interface Requirements

4.1. User Interfaces Overview

High-Level Functionality:

1. Main Menu Screen

- When the user opens the application, the main menu is displayed.
- The user can choose one of the following options:
  - Convert mark to letter grade
  - Calculate GPA using marks
  - Calculate GPA using letter grades

Interaction:

- User selects an option by clicking a button

Feedback:

- The system navigates to the selected feature screen

2. Course Input Screen

- The system asks the user to enter the number of courses
- Based on the number entered, the system dynamically generates input fields

Interaction:

- Typing using keyboard
- Clicking input fields
- Buttons:
  - "Add Course"
  - "Delete Course"
  - "Calculate"

Feedback:

- Input fields appear dynamically
- Deleted courses are removed instantly

- Validation messages appear for invalid inputs

3. Results Screen

- After clicking "Calculate" the system displays:
    - GPA value rounded to two decimal places
    - Converted letter grades if marks were used

Feedback:

- Clear GPA result displayed prominently

4. New Calculation Option

- The user can perform another calculation without restarting

Interaction:

- Button: "Calculate Again" or "Back to Menu"

Feedback:

- The system clears all previously entered **data** and navigates the user back to the main menu

Minimum Interface Requirements:

1. Design Principles

- Simple and user-friendly interface
- Clean layout with clear labels
- Suitable for users with basic computer skills

2. Screen Layout

Each screen should include:

- Title/Header
- Input section
- Action buttons:
  - Calculate
  - Reset / Clear
  - Delete course
- Output/Result section

3. Standard Buttons

The following buttons should be available where applicable:

- Calculate
- Reset
- Add / Delete Course
- Back / Home
- Calculate Again

4. Input Validation & Error Messages

- All inputs must be validated before calculation
- Error messages must be:
  - Clear
  - Simple
  - Displayed near the input field or as a popup

4.2. Hardware Interfaces

1-Supported devices :

The system shall run on standard user devices such as:

- Personal computers (desktop or laptop)
- Mobile devices

2. Input Devices

The system shall accept input through standard hardware devices, including:

- Keyboard: for entering course names, marks, grades, and credit hours
- Mouse or touch input**:** for selecting options and clicking buttons

3. Output Devices

The system shall display results using standard output devices:

- Monitor or screen**:** to display GPA results, grades, and error messages

4.3. Software Interfaces

1. Operating System Interface

The system operates on standard operating systems, including:

- Microsoft Windows
- macOS
- Linux

Purpose:
To provide a runtime environment for executing the application through a web browser

2. Web Browser Interface

The system is accessed through modern web browsers such as:

- Google Chrome
- Microsoft Edge

Purpose:

- To render the graphical user interface (GUI)
- To handle user interactions such as input, button clicks, and displaying results

3. Programming Environment / Runtime

The system depends on the chosen programming language and its runtime environment

Purpose:

- To execute the application logic
- To perform GPA calculations and grade conversions

4. Development Tools Integration

the system interacts with the following tools during development:

- Git & GitHub
  Purpose: Version control, collaboration, and code management
- GitHub Actions (CI Pipeline)
  Purpose: Automated build and testing of the application
- SonarQube
  Purpose: Code quality analysis and maintainability checks

5. Database Interface

There is no database integration in this system.

Constraint:

- All data is entered by the user and stored temporarily in memory during runtime only
- No data is saved or persisted after the session ends

6. Data Input and Output

Input Data:

- Course name
- Credit hours
- Marks or letter grades
- Number of courses

Purpose:
Used by the system to perform GPA calculations and grade conversion.

Output Data:

- Calculated GPA (rounded to two decimal places)
- Converted letter grades (if marks are used)
- Error messages (if invalid input is detected)

Purpose:
To provide results to the user

7. Communication and Data Sharing

- The system does not communicate with external systems
- All processing occurs locally within the application

5. System Features/Modules

5.1 Use cases

Use case 1 : Select Calculation Type

Actor: Student

Description:
The student selects the desired calculation method from the system menu

Main Flow:

1. System displays menu options
2. Student selects one option:

    - Calculate grade from mark
    - Calculate GPA using marks
    - Calculate GPA using grades

Preconditions:

- The system is running successfully
- The main menu is displayed to the student

Postconditions:

- A calculation method is selected
- The system is ready to proceed based on the selected option

Use case 2: Generate Course Input Fields

Actor: System

Description:
The system creates input fields dynamically based on the number of courses

Main Flow:

- System prompts for number of courses
- Student enters number of courses

- System reads number of courses
- System generates corresponding input fields

Postconditions:

- Input fields for all courses are displayed

Use case 3: Delete Course

Actor**:** Student

Description:
The student removes a course from the entered list

Main Flow:

- Student selects a course to delete
- System removes the selected course from the list
- System updates the displayed input fields

Preconditions:

- At least one course has been entered
- Course list is displayed

Postconditions:

- Selected course is removed
- Updated course list is displayed
- Remaining data is ready for processing

Use case 4: Convert Mark to Grade

Actor**:** Student

Description**:**
The system converts numeric marks into letter grades

Main Flow:

- For each course, student enters:

1- Course name
2- Mark

- System reads mark
- Applies grading scale
- Assigns corresponding letter grade
- Show calculated letter grade

Preconditions:

- Valid numeric mark is provided

Postconditions:

- Corresponding letter grade is generated and displayed

Use case 5: Calculate GPA by marks or grades

Actor**:** Student

Description**:** the system will calculate the GPA using the course data either by marks or grades (based on user selection)

Main Flow:

- For each course, student enters:

1-Course name

2-credit hours

3-Mark or grade (based on user selection)

- System prompts user to choose input method
- User select grades
- User enter grades to all courses
- System validates all inputs
- System will calculate the GPA
- System will display the GPA

Alternative path 1 : Using Marks Instead of Grades:

- Using Marks Instead of Grades
- User selects Marks
- User enters marks for all courses
- System validates marks
- System calculates GPA
- System displays GPA

Alternative path 2: Invalid Input:

1-System detects invalid input

2-System displays error message

3-User re-enters correct data

4-Resume from step 1

Preconditions:

- The system is running
- The user has entered at least one course

Each course has:

- Credit hours
- Either grade OR mark

Postconditions:

- GPA is calculated

Use case 6: Perform New Calculation

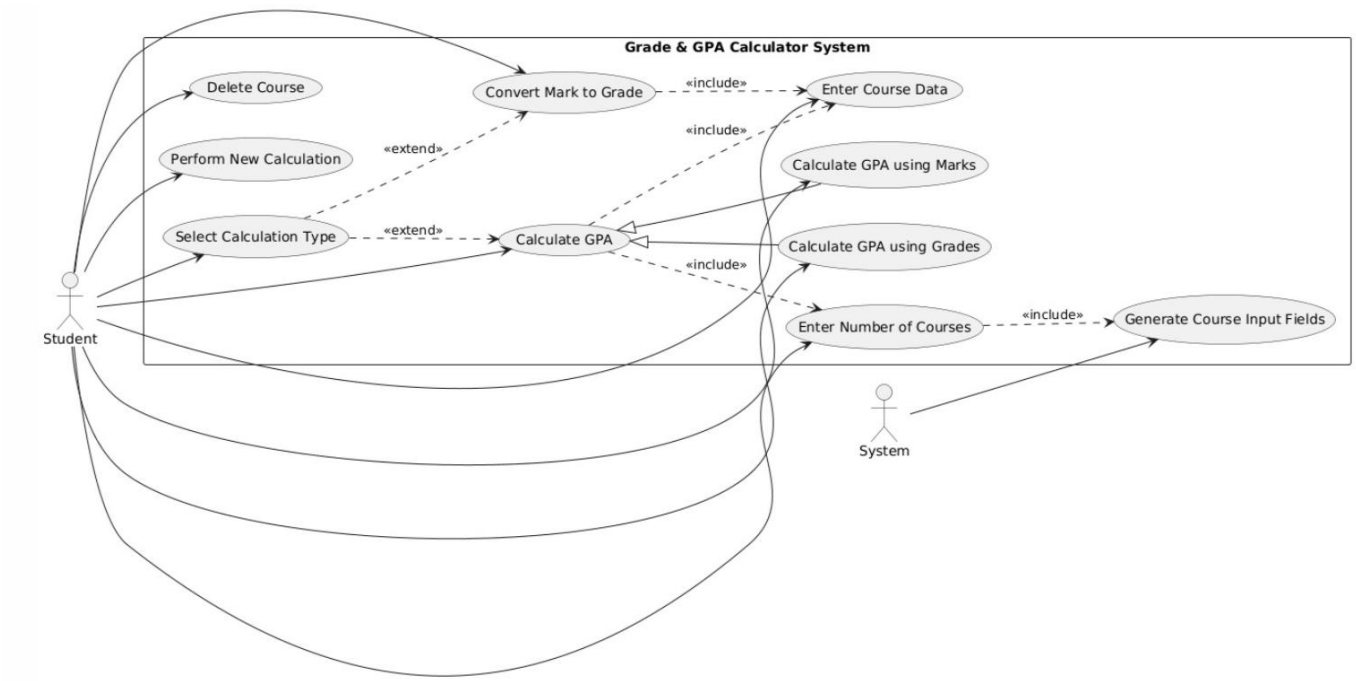
Actor**:** Student

Description**:**
The student performs another calculation without restarting the system

Main Flow:

- System provides option to restart
- Student selects to calculate again
- System returns to main menu

5.2 Use case diagram:



5.3 Functional Requirements

1: The system shall provide a menu that allows the student to choose between calculating a grade from a mark, calculating GPA using marks, or calculating GPA using letter grades

2: The system shall allow the user to enter the number of courses for GPA calculation

3: The system shall dynamically generate input fields for each course based on the number entered

4: The system shall allow the user to enter course details, including course name, credit hours, and either numeric mark or letter grade depending on the selected calculation method

5: the system shall allow the user to delete course

6: The system shall convert numeric marks into corresponding letter grades according to a predefined grading scale

7: The system shall convert letter grades into grade points based on a standard 4.0 GPA scale

8: The system shall calculate the GPA using the entered course data, credit hours, and grade points

9: The system shall display the equivalent letter grade for the marks

10: The system shall present GPA results rounded to two decimal places

11: The system shall allow the user to perform multiple calculations in a single session without restarting the application

6. Nonfunctional Requirements

1- performance requirements

- The system shall calculate GPA and display results within 1 second after user input
- The system shall respond to user inputs immediately without noticeable delay

2-Usability Requirements

- The system shall provide a simple and user-friendly interface suitable for students with basic computer skills.
- The system shall require minimal user interaction to perform calculations

3-Reliability Requirements

- The system shall produce accurate GPA results based on the predefined grading scale
- The system shall maintain consistent behavior during repeated use

4- Availability Requirements

- The system shall maintain an availability of at least **99%**, so it is accessible to users at all times 24/7except for minimal downtime due to maintenance or unforeseen issues

5- Maintainability Requirements

- The system code shall follow clean code principles to ensure readability and maintainability
- The system shall achieve a maintainability rating of A or B using SonarQube